When Dream Decoders Learn the Laboratory: Cross-Dataset Validation and Reproducibility Audit of DREAM Experience Classification

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Abstract

Electroencephalographic classifiers that distinguish reports of dream experience from reports of no experience appear to offer an objective window onto otherwise private mental states. The DREAM database paper reported above-chance discrimination in non-rapid-eye-movement (NREM) and rapid-eye- movement (REM) sleep using pooled, randomly partitioned observations. We audited the released Figure 5 workflow, reconstructed its public inputs, and tested whether the reported signal transfers to unseen participants and unseen laboratories. The code-path dataset comprised 1,065 strict binary reports from five collections (730 NREM; 335 REM). A close adaptation recovered the published scale of performance under record-wise splitting (NREM spectral power, AUROC = 0.569; REM band-filtered Catch22, AUROC = 0.636), but performance collapsed when an entire laboratory was held out (0.493 and 0.470, respectively). In contrast, source-dataset identity was strongly decodable under participant-held-out validation (AUROC = 0.81–0.95). Riemannian covariance features and label-free log-Euclidean domain alignment did not produce reliable transfer. Raw temporal-irreversibility features reached REM AUROC = 0.604 across laboratories, but fell to 0.490 after normalization against phase-randomized surrogates. Participant-cluster bootstrap intervals did not exclude chance consistently in any target laboratory. The released workflow also lacks a required feature module and expects a preformatted Siclari representation absent from all public data versions inspected. These results support within-pool discriminability but not a transportable EEG marker of reported dream experience. Cross-laboratory validation and dataset- identity controls should be treated as required falsification tests for multicenter dream decoding.

1 Introduction

Dream reports create an unusual measurement problem: the target phenomenon is a private experience, while the label becomes available only after awakening. The DREAM database standardized multiple sleep EEG and mentation datasets to enable larger-scale analyses of this problem [1]. In its Figure 5 benchmark, XGBoost classifiers using normalized spectral power and Catch22 time-series features discriminated “Experience” from “No experience” in both NREM and REM sleep. The best reported mean areas under the receiver-operating characteristic curve were 0.586 for NREM spectral power and 0.700 for REM band-filtered Catch22.

These pooled results establish discriminability under the reported validation scheme, but do not by themselves establish transportability. Multicenter EEG datasets differ in hardware, reference, montage, filtering, awakening protocol, class prevalence, and report semantics. A random split can therefore place observations from the same participant and acquisition domain in both training and test sets. Even when participants are separated, a model can use stable laboratory signatures that remain shared across the split. This is a general source of optimistic inference in machine-learning studies [2, 3].

We asked a stricter question: does a classifier trained on other DREAM collections rank reported experience above no experience in a laboratory it has never seen? We first audited the released code and public data. We then implemented a validation ladder that changes only the train-test boundary: random records, held-out participants, and a held-out laboratory. Finally, we tested two mechanistically motivated extensions. Spatial covariance matrices were analyzed on the symmetric-positive-definite manifold, an established EEG transfer-learning framework [4, 5]. Temporal irreversibility was tested because time-directed neural dynamics have been proposed as a signature of conscious state [6]. Both extensions were paired with negative controls designed to expose domain or spectral explanations.

2 Materials and Methods

2.1 Source audit and inclusion

We retrieved the versioned public DREAM collections named by the Figure 5 code and verified every archive against its publisher-provided byte count and MD5 digest. We retained only strict binary labels: DREAM code 2 (Experience) was mapped to 1 and code 0 (No experience) to 0. Experience without recall, ambiguous labels, unknown labels, and wake observations were excluded. Four official EDFs were unavailable for analysis: three files in the Young Adults archive fail CRC despite the archive matching its published MD5, and one Turku EDF is absent from its archive.

The paper describes six collections used across its broader EEG analyses. The released Figure 5 classifier enumerates five. We separately audited the sixth, Oudiette N1Data: its sleeping subset contains 23 strict Experience observations and only one strict No-experience observation, and its Fp1/C3/O1 montage does not match the classifier's F4/C4/O2 montage. It was therefore not inserted post hoc into the classification analysis.

2.2 Reproducibility gaps and explicit adaptations

The deposited workflow could not execute literally against the current public inputs for three reasons. First, it imports and calls `feat_ext_funcs`, which is absent from the code deposit. Second, it declares a semicolon delimiter for De Gennaro Multiple Awakenings, while the public `Records.csv` is comma-separated. Third, it searches for `formatted_*.edf` files with F4-M1, C4-M1, and O2-M1 channels for Siclari, whereas public versions 1–3 contain numerically named high-density channels and no formatted EDFs.

We made the minimum documented adaptations needed for a code-path reproduction. Filtered Catch22 features were computed directly with `pycatch22` [7]. Siclari F4/C4/O2-M1 signals were reconstructed as HydroCel E224/E164/E150 minus E93, the nearest standard HydroCel-256 sensors to F4/C4/O2/M1. Eight records without E93 used the nearest available left-mastoid candidate. These choices make the analysis executable, but prevent a claim of exact numerical reproduction.

2.3 Signal processing and features

We used the final 30 s preceding each report. Signals were resampled to 128 Hz for time-series features. The replication families followed the released workflow: (i) normalized multitaper power from 0.4–35 Hz, summed into delta (0.5–4), theta (4.1–8), alpha (8.1–11), sigma (11.1–15), beta (15.1–20), and gamma (20.1–35) bands across three channels (18 features); (ii) 22 Catch22 features per broadband channel (66 features); and (iii) the same features after each of six band-pass filters (396 features).

For geometric features, each broadband and band-limited epoch was represented by a 3×3 covariance matrix C. We normalized C by its trace, applied one-percent identity shrinkage, and mapped it with the matrix logarithm. The six unique tangent coordinates, with off-diagonal terms multiplied by $\sqrt{2}$, were retained per band and broadband (42 features). A label-free, transductive domain-alignment variant subtracted the log-Euclidean mean of each laboratory and stage before classification.

Temporal irreversibility was summarized per channel at lags 1, 2, 4, and 8 by (i) Jensen-Shannon divergence between forward and reversed ordinal-pattern distributions of embedding dimension three and (ii) a signed cubic time-asymmetry statistic. To separate nonlinear temporal ordering from spectral effects, we generated 24 deterministic phase-randomized surrogates per channel and expressed each statistic relative to its surrogate distribution.

2.4 Classifier and validation ladder

All binary models used the released XGBoost settings [8]: learning rate 0.1, depth 3, subsampling 0.8, feature subsampling 0.8, and the workflow's implicit default of ten boosting rounds. No feature-family-specific hyperparameter search was performed.

Record-wise and participant-held-out results used five folds and 20 shuffled, deterministic repetitions. Participant separation used stratified group cross-validation. Laboratory-held-out analysis trained on all other collections and evaluated the untouched collection once, separately for NREM and REM. Because Siclari contributes only NREM and Turku only REM, each stage has four eligible target laboratories. The primary cross-laboratory summary is the unweighted mean of target-laboratory AUROCs. We also report a version in which each target participant has equal metric weight. Uncertainty intervals were obtained from 2,000 participant-cluster bootstrap samples within each target laboratory.

As a negative control, we predicted dataset identity with the same features using participant-held-out folds and macro one-versus-rest AUROC. High dataset decodability indicates residual domain information but does not, on its own, prove that every experience prediction is artifactual.

3 Results

3.1 Random-split performance is approximately recoverable

The adapted workflow recovered the scale and direction of the published benchmark. NREM spectral power reached mean record-wise AUROC = 0.569, compared with 0.586 reported in DREAM. REM band-filtered Catch22 reached 0.636, below the reported 0.700, as expected given the absent feature module and Siclari preprocessing. Holding out participants caused only modest changes and did not expose the principal failure mode (Table 2).

3.2 Performance collapses under laboratory shift

The published best-performing families did not transfer. NREM PSD fell from 0.569 under record splitting to 0.493

Table 1: Code-path datasets included in experience classification. Counts are after label, stage, and file-integrity exclusions. Subject identifiers are scoped to their source collection.

Collection	Records	Subjects	No exp.	Experience	NREM	REM
Zhang & Wamsley 2019	212	28	56	156	186	26
De Gennaro Multiple Awakenings	435	19	121	314	281	154
De Gennaro Young Adults	62	62	24	38	32	30
Tononi/Siclari Serial Awakenings	231	39	107	124	231	0
REM Turku	125	18	3	122	0	125
Total	1,065	166	311	754	730	335

Table 2: Validation ladder for reported-experience classification. Random-record and new-subject values average five folds within each of 20 repetitions. New- laboratory values are macro-averages across four target collections. The last column gives the same laboratory-held-out ranking with equal total weight per target participant.

Stage	Feature family	Random record	New subject	New laboratory	New lab., subject-equal
NREM	PSD	0.569	0.550	0.493	0.484
NREM	Catch22 filtered	0.574	0.555	0.441	0.444
NREM	Riemannian covariance	0.560	0.522	0.502	0.480
NREM	Surrogate-normalized irreversibility	0.554	0.548	0.520	0.512
REM	PSD	0.596	0.593	0.481	0.497
REM	Catch22 broadband	0.639	0.616	0.573	0.561
REM	Catch22 filtered	0.636	0.663	0.470	0.455
REM	Riemannian covariance	0.642	0.640	0.544	0.523
REM	Log-Euclidean alignment	0.617	0.616	0.559	0.565
REM	Temporal irreversibility	0.599	0.612	0.604	0.589
REM	Surrogate-normalized irreversibility	0.608	0.628	0.490	0.464

across laboratories. REM filtered Catch22 rose under participant separation (0.663) yet fell to 0.470 when the acquisition domain changed. Subject-equal estimates were 0.484 and 0.455, respectively. At a probability threshold of 0.5, subject-weighted balanced accuracy was approximately 0.5 for the REM cross-laboratory tests, showing that calibration did not transport even when AUROC modestly exceeded chance.

Target-specific uncertainty was substantial. No candidate produced a favorable cluster-bootstrap interval excluding 0.5 in every target. REM temporal irreversibility, for example, yielded subject-equal AUROC and 95% intervals of 0.725 [0.500, 0.910] in Young Adults, 0.420 [0.315, 0.536] in Multiple Awakenings, 0.568 [0.280, 0.860] in Zhang, and 0.642 [0.363, 0.874] in Turku. The Turku estimate is especially fragile because only three No-experience observations are available.

3.3 Dataset identity dominates the feature space

Dataset identity was strongly decodable despite participant-held-out evaluation (Table 3). Catch22 produced macro AUROC up to 0.940 in NREM and 0.953 in REM. Riemannian covariance also retained strong domain information. Log-Euclidean centering reduced NREM identity decoding from 0.918 to 0.770, but did not reduce REM identity decoding materially and improved REM experience transfer only from 0.544 to 0.559.

Table 3: Participant-held-out dataset-identity AUROC.

Feature family	NREM	REM
PSD	0.839	0.808
Catch22 broadband	0.930	0.953
Catch22 filtered	0.940	0.912
Riemannian covariance	0.918	0.845
Log-Euclidean alignment	0.770	0.842

3.4 Temporal asymmetry does not survive spectral control

Unnormalized temporal-irreversibility features were the strongest exploratory REM family under laboratory shift (AUROC = 0.604 by record; 0.589 with equal participant weight). This advantage disappeared after normalization against phase-randomized surrogates (0.490 and 0.464). The surrogate procedure preserves each channel's spectrum while disrupting nonlinear temporal order. The reduction therefore argues against a domain-invariant arrow-of-time signal in these features and favors

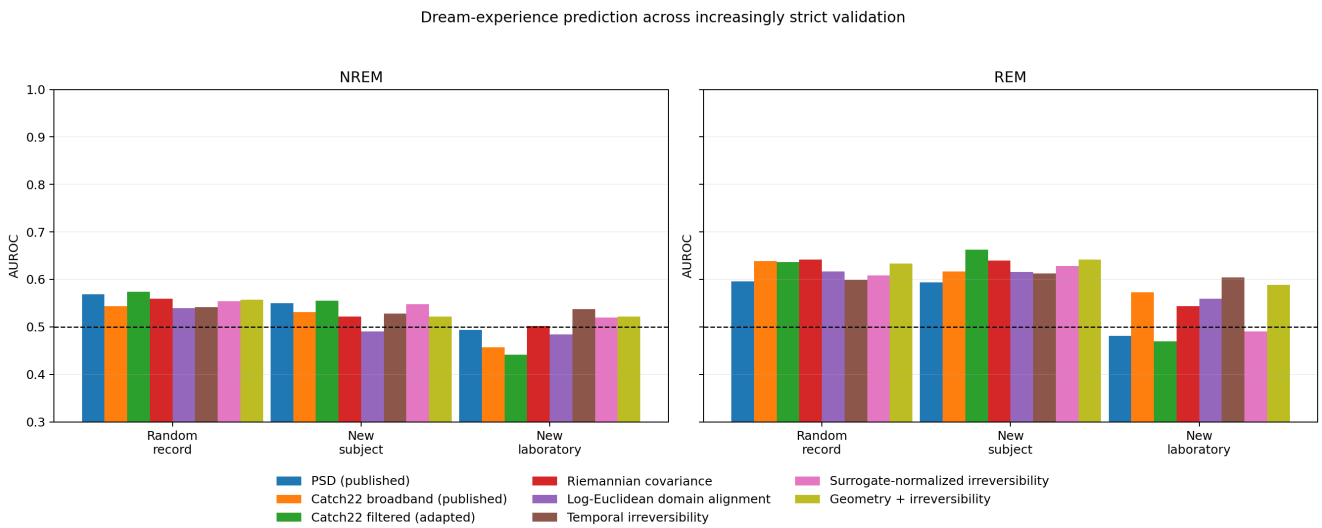

Figure 1: Experience-prediction AUROC across increasingly strict validation. The dashed line marks chance ranking. Random-record and new-subject estimates retain above-chance REM performance for several families; withholding a whole laboratory produces large, method-dependent collapses. Bars show point estimates and should not be interpreted as inferential intervals.

spectral or acquisition-related explanations.

4 Discussion

This reanalysis distinguishes three claims that can otherwise be conflated. First, DREAM EEG contains information that discriminates report labels under a pooled random split; the approximate reproduction supports that claim. Second, some of this information persists when participants are separated within the same collection. Third, the resulting decision rule transports to a new laboratory. The present evidence does not support the third claim.

The contrast between participant-held-out and laboratory-held-out performance is particularly informative. REM filtered Catch22 improved after participant separation yet inverted under domain shift. Thus, participant-level leakage is not the only concern. Stable protocol and instrumentation signatures can remain available on both sides of a participant-held-out split. The dataset-identity control confirms that these signatures are large relative to the experience signal.

Advanced mathematics did not automatically solve the domain problem. Riemannian covariance is well motivated for multichannel EEG, but only three common channels were available and their references were heterogeneous. Label-free log-Euclidean centering removed a portion of the NREM site mean, not the broader distributional differences. Likewise, temporal irreversibility had an appealing theoretical connection to conscious dynamics, but the phase- surrogate result showed why a targeted null model is essential before giving a nonlinear feature a mechanistic interpretation.

The result should not be read as evidence that EEG contains no correlates of dreaming. It shows that the tested representations and released workflow do not yet establish a universal marker of a reported experience. Report labels also remain imperfect proxies: "No experience" can combine absent experience, failed encoding, forgetting, and protocol-specific categorization. Cross-dataset failure may therefore reflect genuine construct heterogeneity as well as measurement shift.

4.1 Limitations

This was an exploratory reanalysis rather than a prospective preregistration. The missing feature module and unavailable preformatted Siclari files required adaptations, and the filtered Catch22 result is not an exact reproduction. Laboratory-held-out inference rests on four target domains per sleep stage, with extreme REM imbalance in Turku. Only the common final 30 s and three channels were analyzed. The domain-alignment analysis used the unlabeled target batch and is therefore transductive. Our literature search did not locate an earlier laboratory-held-out validation of this exact DREAM classifier, but it was not a registered systematic review; novelty should be confirmed before submission.

5 Conclusion

The DREAM Figure 5 signal is approximately reproducible under record-wise validation but is not reliably transportable across laboratories. Dataset identity is substantially easier to decode than reported dream experience, and neither Riemannian alignment nor phase-controlled temporal irreversibility provides a robust rescue. Multicenter dream-decoding studies should report a validation ladder, dataset-identity negative controls, participant-clustered uncertainty, and target-laboratory performance before interpreting pooled accuracy as a general neural marker of private experience.

Data and code availability

All source collections are public through the DREAM database and versioned Figshare records. The analysis workspace records source URLs, file sizes, and MD5 digests; scripts and fold-level outputs are frozen in local commit `ebbaf97`. The paper-only public repository contains the LaTeX source, bibliography, figure, and PDF, separate from raw and expanded EEG files. Exact numerical replication also requires the unreleased `feat_ext_funcs` implementation and the preformatted Siclari files expected by the deposited workflow.

Ethics statement

This study reanalyzed de-identified, publicly released data and did not recruit participants or modify source records. Ethical approvals and consent procedures are described in the originating studies and DREAM data records.

Competing interests

The author declares no competing interests.

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